1 Background

Soft magnetic cores are important components in many inductors and transformers, where they provide a path for magnetic flux and influence electromagnetic performance [1]. In power converters, inductors and transformers can occupy a substantial fraction of the system volume, motivating research into more compact designs [2]. However, flux density is not always distributed uniformly within a magnetic core. Depending on the core geometry, material properties and operating conditions, this nonuniformity can lead to uneven core losses and localised heating [3]. In some inductor designs, regions near the conductor may approach saturation while other regions remain underutilised [4].

Flux distribution can be controlled through the core geometry and winding arrangement. For example, Cui et al. designed an inductor with an improved flux distribution by shaping the core and distributing the winding turns [5]. Another approach is to vary permeability spatially within the core, using regions with different magnetic properties to tailor the flux distribution [4, 6]. Wang et al. experimentally demonstrated that a three-permeability inductor achieved higher inductance than a single-permeability counterpart across the tested load range and improved the efficiency of the tested high-frequency DC/DC converter [4]. Liu et al. also reported a smaller core volume at the same inductance for a three-permeability design compared with a single-permeability core [6]. These studies demonstrate the potential benefits of spatial permeability variation for particular designs, rather than establishing that it is always preferable to geometry optimisation.

Manufacturing cores with spatially varying permeability requires control over the distribution of magnetic material properties. Additive manufacturing offers one approach because different material formulations can be deposited in selected regions of a component. Liu et al. demonstrated this using a multi-extruder paste-extrusion 3D printer to fabricate toroidal cores with three distinct permeability regions [6]. Continuous radial permeability variation has also been demonstrated by imposing a temperature gradient during sintering [7]. More recently, Lu et al. produced discrete permeability regions by varying compaction pressure during core formation, illustrating that several manufacturing approaches to multipermeability cores are possible [8].

The choice of feedstock for material extrusion involves a trade-off between magnetic performance and manufacturability. Producing filament-based composites with high magnetic-particle loading while retaining suitable processability can be challenging [9]. In their review of soft magnetic feedstocks, Cañada et al. report that filament-based cores in the studies they surveyed generally exhibited relative permeabilities below 10. Paste-based feedstocks can achieve higher permeabilities, but their rheology can complicate dimensional accuracy, and some ferrite formulations require high-temperature sintering. For example, the ferrite paste investigated by Liu et al. required sintering above 900 °C, whereas their powdered-iron paste could be cured below 250 °C [6]. In comparison, Cañada et al. demonstrated pellet-extruded nylon-12/FeSiAl composite cores with a relative permeability of 42 without high-temperature post-processing [10]. These values come from different material systems and studies, so they should not be interpreted as a direct, controlled comparison of the three feedstock types.

The composition of a polymer-based magnetic composite is another factor that can influence its magnetic properties. Hu et al. experimentally investigated the permeability of polymer composites containing different magnetic inclusions, although their matrix material differed from the nylon-12 composites considered here [11]. Huber et al. demonstrated that magnetic composition could be varied during filament-based printing by blending NdFeB-filled PA12 filament with unfilled PA12 filament. Their method controlled the remanent magnetisation distribution of permanent magnets to produce a predefined stray field, rather than controlling the effective permeability of a soft-magnetic core [12].

Composition transitions have also been demonstrated using pellet-fed extrusion in non-magnetic material systems. Brackett et al. studied in-situ blending of neat and carbon-fibre-filled ABS pellets and found that material transitions occupied a finite distance and depended on the transition direction [13]. Elsersawy et al. further showed that the mixing geometry within a pellet extruder affected the uniformity of blended PLA materials [14]. These studies identify process considerations relevant to composition-controlled extrusion, but do not establish how pellet blend ratio affects the effective permeability of a soft-magnetic composite.

The literature reviewed here therefore establishes that pellet-based material extrusion can produce soft-magnetic cores, that magnetic composition can be varied during filament-based printing of permanent magnets, and that pellet-fed systems can blend materials and produce finite composition transitions. However, it remains to be established whether blending magnetic-filled pellets with unfilled pellets of a compatible polymer matrix provides a predictable and reproducible way to control the effective permeability of a pellet-extruded soft-magnetic composite. This project will first investigate the relationship between pellet blend ratio and the effective permeability of the printed material. If this relationship is sufficiently controllable, subsequent work can investigate changing the feed ratio during printing to produce distinct permeability regions in a core.

2 Problem Statement

2.1 Research Question

To address the identified problem, the research question is defined as follows:

2.2 Requirements

To address the research question and capture the overarching scientific plan, the following requirements have been defined:

2.3 Desirables

Beyond the essential requirements, additional desirable outcomes are below:

2.4 Delimitations

The project will employ commercially available additive manufacturing equipment, modified for the task at hand.

3 Methodology

3.1 Requirements

To address this research question, the following requirements have to be met:

3.2 Desirables

Beyond the essential requirements, additional desirable outcomes are below:

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